



针对星地通信终端的逆向分析及智能仿真技术

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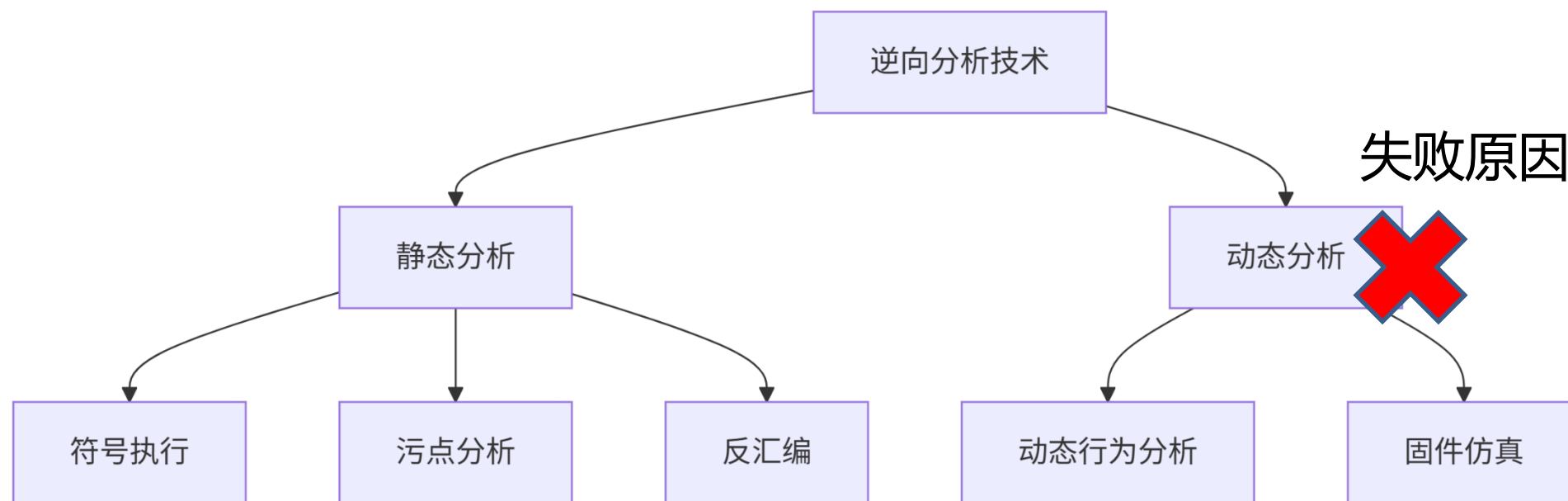
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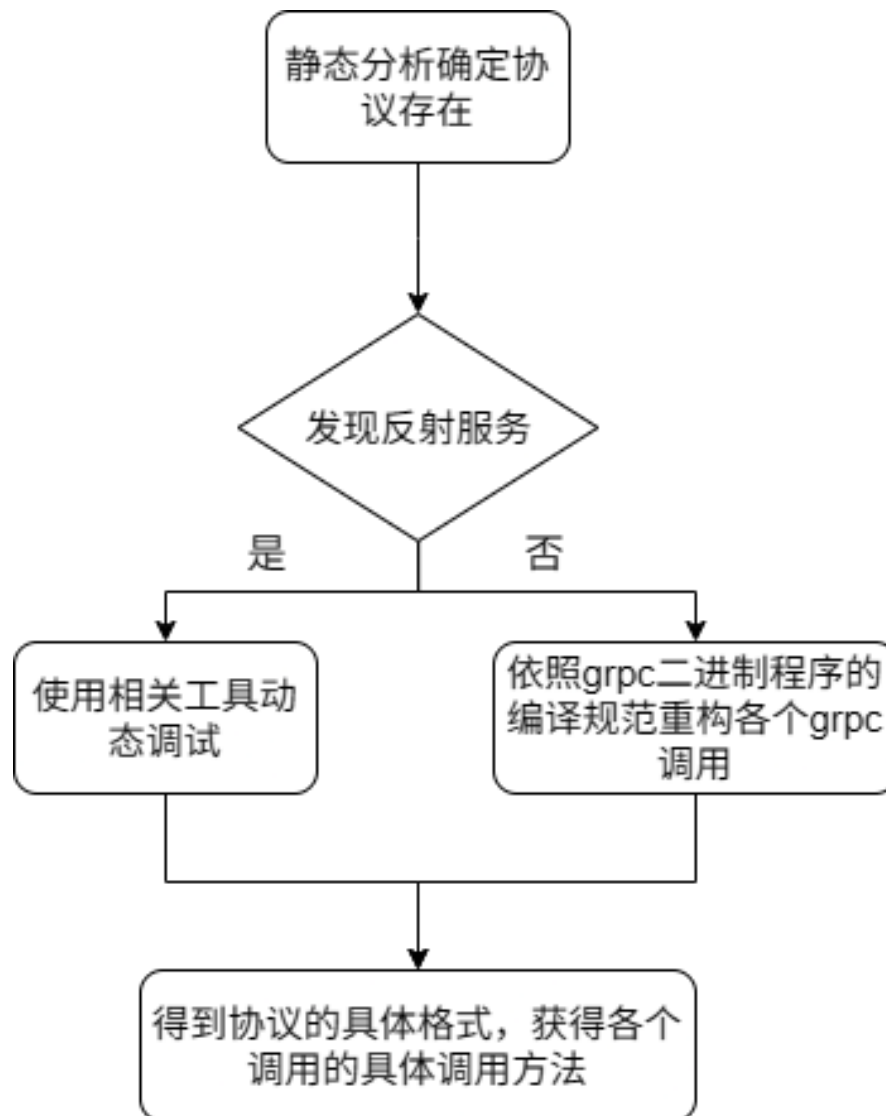
星地通信终端作为卫星互联网的核心嵌入式设备，在国防、通信等关键领域发挥着不可替代的作用。研究并分析这类设备的安全漏洞，对于保障国家空间基础设施安全具有重大战略意义。然而，由于星地终端特殊的硬件架构和各种硬件依赖，较传统IoT安全研究来说，严重限制了其通过仿真进行动态调试的效率，这种研究空白既带来了挑战，也为安全领域提供了宝贵的研究机遇。

逆向分析技术 = 静态分析技术 + 动态分析技术



失败原因：复杂的硬件依赖

Grpc协议服务端逆向方法



嵌入式设备系统级仿真技术

维度	Qemu-system	Renode
适用层级	文件系统	Bootloader/内核/芯片级
外设模拟	较难	简单
性能	高	较低
典型场景	文件系统内软件分析	定制化硬件/固件底层逆向

动态仿真环境搭建



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根据一般设计规律从镜像中分离各个部分，并使用fiptool解压各个镜像二进制文件

```
luheng@luheng-virtual-machine: /mnt/workspace/Baid...  
00000000: 0100 64aa 7856 3412 0000 0000 5555 0000 ..d.xV4.....UU..  
00000010: d6e2 69ea 5d63 e411 8d8c 9fba be99 56a5 ..i.]c.....V..  
00000020: 8800 0000 0000 0000 9000 0000 0000 0000 .....  
00000030: 0000 0000 0000 0000 5ff9 ec0b 4d22 3e4d ....._..M">M  
00000040: a544 c39d 81c7 3f0a 1801 0000 0000 0000 .D....?.....  
00000050: 78f1 0000 0000 0000 0000 0000 0000 0000 x.....  
00000060: 0000 0000 0000 0000 0000 0000 0000 0000 .....  
00000070: 90f2 0000 0000 0000 0000 0000 0000 0000 .....  
00000080: 0000 0000 0000 0000 6f3b f58f 941b 28e5 .....o;....(  
00000090: bcec 89dc d196 00e4 630e 9070 28b9 1e3b .....c..p(..  
000000a0: e89a c4e1 df59 71b1 1983 8b93 2884 405b .....Yq.....(.[  
000000b0: 697b da39 c75a 4f04 f5e0 a667 5953 ae16 i{.9.Z0....gYS..  
000000c0: ec07 66da 7b74 b10c 5354 4d31 0100 0000 ..f.{t..STM1....  
000000d0: 0204 0000 0001 0000 8508 f7ba 1862 b11b .....b..  
000000e0: 4b81 8e63 3eb1 7bd8 8f45 e7f4 6028 8c7e K..c>.{..E..`(.~  
000000f0: 3b8f 7bc5 d075 f2aa af12 ba1f 332f 5dd4 ;.{..u.....3/].  
00000100: f411 58af ddae 7648 043b 3878 f151 6342 ..X...vH.;8x.QcB  
00000110: 1230 6a8a 8e0a 2027 f403 01aa e03f 0710 .0j... '.....?  
00000120: 00c0 18d5 df3f 03d5 ff44 03d5 4101 82d2 .....?...D..A..  
00000130: 0010 38d5 0000 01aa 0010 18d5 df3f 03d5 ..8.....?..  
00000140: c07e 0710 c1ab 0910 2100 00cb 7a0b 0094 ..~.....!...z...  
00000150: 0001 0058 2101 0058 840b 0094 470b 0094 ...X!..X....6...  
00000160: e003 14aa ec08 0094 910b 0094 0000 0000 .....  
:
```

```
luheng@luheng-virtual-machine:/mnt/workspace/BaiduSyncdisk/renode-project/sl_router$ fiptool info ./boot1_0x0  
Trusted Boot Firmware BL2: offset=0x118, size=0xF178, cmdline="--tb-fw"  
Trusted Boot Firmware BL2 certificate: offset=0x88, size=0x90, cmdline="--tb-fw-cert"  
luheng@luheng-virtual-machine:/mnt/workspace/BaiduSyncdisk/renode-project/sl_router$ fiptool info ./uboot1_600000  
Trusted Boot Firmware BL2: offset=0x230, size=0x2BAC0, cmdline="--tb-fw"  
EL3 Runtime Firmware BL31: offset=0x2BEA0, size=0x1B9B0, cmdline="--soc-fw"  
Non-Trusted Firmware BL33: offset=0x47970, size=0x47778, cmdline="--nt-fw"  
Trusted key certificate: offset=0x2BCF0, size=0x90, cmdline="--trusted-key-cert"  
SoC Firmware key certificate: offset=0x2BD80, size=0x90, cmdline="--soc-fw-key-cert"  
Non-Trusted Firmware key certificate: offset=0x47850, size=0x90, cmdline="--nt-fw-key-cert"  
Trusted Boot Firmware BL2 certificate: offset=0x1A0, size=0x90, cmdline="--tb-fw-cert"  
SoC Firmware content certificate: offset=0x2BE10, size=0x90, cmdline="--soc-fw-cert"  
Non-Trusted Firmware content certificate: offset=0x478E0, size=0x90, cmdline="--nt-fw-cert"
```

动态仿真环境搭建



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使用IDA逆向，获得相关信息，搭建基本的Renode仿真环境

```
ROM:0000000000000000 ; __int64 sub_0()
ROM:0000000000000000 sub_0 ; DATA XREF: sub_1E20+94+o
ROM:0000000000000000 ; sub_1E20+AC+o ...
ROM:0000000000000000 ; FUNCTION CHUNK AT ROM:0000000000000178 SIZE 00000004 BYTES
ROM:0000000000000000
ROM:0000000000000000 B loc_28
ROM:0000000000000000 ; -----
ROM:0000000000000004 ALIGN 8
ROM:0000000000000008 off_8 DCQ 0xBF000000 ; DATA XREF: sub_1E20+A4+r
ROM:0000000000000008 ; sub_E1D8+190+r ...
ROM:0000000000000010 off_10 DCQ qword_46A78 ; DATA XREF: sub_E1D8+B0+r
ROM:0000000000000010 ; sub_E1D8+180+r ...
ROM:0000000000000018 off_18 DCQ qword_46A78 ; DATA XREF: sub_E1D8+98+r
ROM:0000000000000018 ; sub_E1D8+474+r ...
ROM:0000000000000020 DCQ 0x4C498
ROM:0000000000000028 ; -----
ROM:0000000000000028 loc_28 B ; CODE XREF: sub_0+j
ROM:0000000000000028 ; -----
ROM:000000000000002C loc_2C B ; CODE XREF: sub_0:loc_178+j
ROM:000000000000002C ADR X0, sub_1800
ROM:0000000000000030 MRS X1, CurrentEL
ROM:0000000000000034 CMP X1, #0xC
ROM:0000000000000038 B.EQ loc_4C
ROM:000000000000003C CMP X1, #8
ROM:00000004 0000000000000004: sub_0+4 (Synchronized with Hex View-1)
```

通过指针地址分析确定了代码段基址为
0xBF000000

厚德弘毅 求是笃行

```
// APU
apu0: CPU.ARMv8A @ cluster0
cpuType: "cortex-a53"
cpuId: 0
genericInterruptController: apuGic

apu0Timer: Timers.ARM_GenericTimer @ apu0
frequency: 100000000
NonSecureEL2PhysicalTimerIRQ -> apuGic#0@29
EL1PhysicalTimerIRQ -> apuGic#0@30
EL1VirtualTimerIRQ -> apuGic#0@27
EL3PhysicalTimerIRQ -> apuGic#0@26

apuGic: IRQControllers.ARM_GenericInterruptController @ {
sysbus new Bus.BusMultiRegistration { address: 0x26110000; size: 0x1000; region: "distributor" };
sysbus new Bus.BusMultiRegistration { address: 0x26120000; size: 0x11000; region: "cpuInterface" }
}
[0-3] -> apu0@[0-3]
[4-7] -> apu1@[0-3]
[8-11] -> apu2@[0-3]
[12-15] -> apu3@[0-3]

baseRAM: Memory.MappedMemory @ sysbus 0x0
size: 0x100000000
```

```
:name: boot!
$uboot?=@uboot1_extracted/nt-fw.bin
include @log_mem.cs
using sysbus
mach create $name
machine LoadPlatformDescription @bootloader.repl
machine StartGdbServer 3333 true sysbus.apu0
macro reset
""
sysbus LoadBinary $uboot 0xBF000000
apu0 PC 0xBF000000
""
runMacro $reset
```

观察MMIO寄存器的访问，判断外设模型

```
16:19:46.0249 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0250 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0x20
16:19:46.0250 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0251 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0x43
16:19:46.0251 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0252 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0x50
16:19:46.0252 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0253 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0x55
16:19:46.0254 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0255 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0x20
16:19:46.0255 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0256 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0x2E
16:19:46.0256 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0257 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0x2E
16:19:46.0258 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0258 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0x2E
16:19:46.0259 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0260 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0xD
16:19:46.0260 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0261 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0xA
16:19:46.0262 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0262 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0xD
16:19:46.0263 [WARNING] sysbus: [apu0: 0xBFFBDFBC] ReadDoubleWord from non existing peripheral at 0x8850014.
16:19:46.0264 [WARNING] sysbus: [apu0: 0xBFFBE010] WriteDoubleWord to non existing peripheral at 0x8850004, value 0xA
```

都是ASCII码输出，判断
很有可能是UART设备，
数据寄存器地址为
0x8850004

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观察MMIO寄存器的访问地址，配合静态分析和gdb动态分析，逆向MMC设备接口

```
16:37:39.2273 [WARNING] sysbus: [apu0: 0xbffbd970] ReadDoubleWord from non existing peripheral at 0x0.
16:37:39.2273 [WARNING] sysbus: [apu0: 0xbffbd970] WriteDoubleWord to non existing peripheral at 0x1E301800, value 0x0.
16:37:39.2273 [WARNING] sysbus: [apu0: 0xbffbd93c] ReadDoubleWord from non existing peripheral at 0x1E400004.
16:37:39.2274 [WARNING] sysbus: [apu0: 0xbffbd93c] WriteDoubleWord to non existing peripheral at 0x1E400004, value 0x2.
16:37:39.2274 [WARNING] sysbus: [apu0: 0xbffbfef90] ReadDoubleWord from non existing peripheral at 0x1E400004.
16:37:39.2275 [WARNING] sysbus: [apu0: 0xbffbfef90] WriteDoubleWord to non existing peripheral at 0x1E400004, value 0x20.
16:37:39.2275 [WARNING] sysbus: [apu0: 0xbffbfef90] ReadDoubleWord from non existing peripheral at 0x1E400004.
16:37:39.2275 [WARNING] sysbus: [apu0: 0xbffbfef90] WriteDoubleWord to non existing peripheral at 0x1E400004, value 0x200.
16:37:39.2275 [WARNING] sysbus: [apu0: 0xbffbfef90] ReadDoubleWord from non existing peripheral at 0x1E400004.
16:37:39.2276 [WARNING] sysbus: [apu0: 0xbffbfef90] WriteDoubleWord to non existing peripheral at 0x1E400004, value 0x2000.
16:37:39.2276 [WARNING] sysbus: [apu0: 0xbffbfef90] ReadDoubleWord from non existing peripheral at 0x1E400004.
16:37:39.2277 [WARNING] sysbus: [apu0: 0xbffbfef90] WriteDoubleWord to non existing peripheral at 0x1E400004, value 0x20000.
16:37:39.2278 [WARNING] sysbus: [apu0: 0xbffbfef90] ReadDoubleWord from non existing peripheral at 0x1E400004.
16:37:39.2278 [WARNING] sysbus: [apu0: 0xbffbfef90] WriteDoubleWord to non existing peripheral at 0x1E400004, value 0x200000.
16:37:39.2278 [WARNING] sysbus: [apu0: 0xbffbfef90] ReadDoubleWord from non existing peripheral at 0x1E400004.
16:37:39.2279 [WARNING] sysbus: [apu0: 0xbffbfef90] WriteDoubleWord to non existing peripheral at 0x1E400004, value 0x2000000.
16:37:39.2280 [WARNING] sysbus: [apu0: 0xbffbfef90] ReadDoubleWord from non existing peripheral at 0x1E400000.
16:37:39.2280 [WARNING] sysbus: [apu0: 0xbffbfef90] WriteDoubleWord to non existing peripheral at 0x1E400000, value 0x20000000.
16:37:39.2281 [WARNING] sysbus: [apu0: 0xbffbfef90] ReadDoubleWord from non existing peripheral at 0x1E400000.
16:37:39.2281 [WARNING] sysbus: [apu0: 0xbffbd9c8] ReadDoubleWord from non existing peripheral at 0x1E400140.
16:37:39.2281 [WARNING] sysbus: [apu0: 0xbffbd9c8] WriteDoubleWord to non existing peripheral at 0x1E400140, value 0x3FE000.
16:37:39.2282 [WARNING] sysbus: [apu0: 0xbffbd9c8] WriteDoubleWord to non existing peripheral at 0x1E300400, value 0x1001101.
16:37:39.2282 [WARNING] sysbus: [apu0: 0xbffbd9c8] WriteDoubleWord to non existing peripheral at 0x1E300404, value 0x111100C8.
16:37:39.2282 [WARNING] sysbus: [apu0: 0xbffbd9c8] WriteDoubleWord to non existing peripheral at 0x1E300408, value 0x10110111.
16:37:39.2282 [WARNING] sysbus: [apu0: 0xbffbd9c8] WriteDoubleWord to non existing peripheral at 0x1E30040C, value 0x101.
16:37:39.2283 [WARNING] sysbus: [apu0: 0xbffbd9c8] WriteDoubleWord to non existing peripheral at 0x1E300410, value 0x10F.
16:37:39.2283 [WARNING] sysbus: [apu0: 0xbffbd9c8] WriteWord to non existing peripheral at 0x1E30000C, value 0x0.
16:37:39.2283 [WARNING] sysbus: [apu0: 0xbffbd9c8] WriteWord to non existing peripheral at 0x1E30000E, value 0x0.
16:37:39.2284 [WARNING] sysbus: [apu0: 0xbffbd6a0] ReadDoubleWord from non existing peripheral at 0x1E300040.
16:37:39.2284 [WARNING] sysbus: [apu0: 0xbffbd77c] ReadWord from non existing peripheral at 0x1E3000FE.
16:37:39.2308 [WARNING] usart1: Unhandled read from offset 0x14. (64)
16:37:39.2309 [INFO] usart1: MMC: sdhci_setup_cfg: Hardware doesn't specify base clock frequency
```

```
ROM:00000000BFFBCD20 ;-----
ROM:00000000BFFBCD20 loc_BFFBCD20
ROM:00000000BFFBCD20 ; CODE XREF: mmc
ROM:00000000BFFBCD20 LDR W0, [X24]
ROM:00000000BFFBCD20 CMP W0, W25
ROM:00000000BFFBCD20 B.HI loc_BFFBCD5C
ROM:00000000BFFBCD20 MOV W1, W26
ROM:00000000BFFBCD20 MOV W2, W27
ROM:00000000BFFBCD30 ADD X0, X28, #aSMmcDBusy_0@PAA
ROM:00000000BFFBCD30 BL printf_string
ROM:00000000BFFBCD30 W1, [X24]
ROM:00000000BFFBCD30 LSL W1, W1, #1
ROM:00000000BFFBCD44 CMP W1, #0xc80
ROM:00000000BFFBCD44 loc_BFFBCF28
ROM:00000000BFFBCD44 B.HI X0, aTimeoutIncrease_0 ;
ROM:00000000BFFBCD54 W1, [X24]
ROM:00000000BFFBCD54 STR BL printf_string
ROM:00000000BFFBCD5C ; CODE XREF: mmc
ROM:00000000BFFBCD5C ADD W25, W25, #1
ROM:00000000BFFBCD60 MOV X0, #0x3E8
ROM:00000000BFFBCD60 BL wait
ROM:00000000BFFBCD68 loc_BFFBCD68 ; CODE XREF: mmc
ROM:00000000BFFBCD68 LDR X0, [X19, #8]
ROM:00000000BFFBCD68 LDR W23, [X0, #0x24]
ROM:00000000BFFBCD70 DMB SY
ROM:00000000BFFBCD70 ANDS W23, W22, W23
ROM:00000000BFFBCD78 B.NE loc_BFFBCD20
ROM:00000000BFFBCD7C DMB SY
ROM:00000000BFFBCD80 LDR X0, [X19, #8]
ROM:00000000BFFBCD84 MOV W1, #0xFFFFFFFF
ROM:00000000BFFBCD88 STR W1, [X0, #0x30]
ROM:00000000BFFBCD8C LDRH W3, [X21]
ROM:00000000BFFBCD90 CMP W3, #0x13
ROM:00000000BFFBCD94 CSET W3, EQ
ROM:00000000BFFBCD98 CMT W3, #0x15
ROM:00000000BFFBCD9C W? W? W?D NE
00BFFBCD0 00000000BFFBCD80: mmc interface+0 (Synchronized with Hex View-1. Pa...
```

编写相关接口仿真代码，正确搭建外设模型

```
private void DefineRegisters()
{
    Register.RData.Define(this, name: "USART_SR")
        .WithValueField(0, 32, writeCallback: (_, value) =>
        {
            //this.Log(LogLevel.Warning, "Trying to transmit a character, but the transmitter is busy");
            CharReceived?.Invoke((byte)value);
            transmissionComplete.Value = true;
            uart_message+=(char)value;
            if ((char)value == '\n'){
                this.Log(LogLevel.Info, uart_message);
                uart_message = "";
            }

            Update();
        }
        , valueProviderCallback: _ =>
        {
            uint value = 0;
            var cpu = sysbus.GetCurrentCPU();
            var pc = cpu.PC;
            this.Log(LogLevel.Info, "{0:X} provided data", pc);
            // "Cleared by a USART_SR register followed by a read to the USART_DR register."
            // We can assume that USART_SR has already been read on the ISR.
            idleLineDetected.Value = false;
            //this.Log(LogLevel.Warning, "RPT");
            if(receiveFifo.Count > 0)
            {
                value = (uint)receiveFifo.Dequeue();
            }
            readFifoNotEmpty.Value = receiveFifo.Count > 0;
            Update();
            return value;
        }
    );
}
```

```
else if(offset == 0x30){
    //this.Log(LogLevel.Info, "[cpu: {0:X}] Setting offset 0x30: 0x{1:X}.", pc, result);
}
else if(offset == 0x24){
    //result = 0x0C00;
    result = 0x49e5b74;
    //this.Log(LogLevel.Info, "[cpu: {0:X}] Setting offset 0x24: 0x{1:X}.", pc, result);
}
else if(offset == 0x20){
    if(this.last_read_addr + this.block_size == this.addr){
        if(this.count == 0){
            this.Log(LogLevel.Info, "Error! Exceeding read count!");
        }
        else{
            this.mmc_array = this.ReadFileSegment("/mnt/workspace/BaiduSyncdisk/renode-p", this.addr, (int)this.block_size);
            this.count--;
            this.last_read_addr = this.addr;
        }
    }
    else if(this.last_read_addr + this.block_size < this.addr){
        this.Log(LogLevel.Info, "Error! Abrupt Change in addr detected!");
    }
    result = BitConverter.ToUInt32(this.mmc_array, (int)(this.addr - this.last_read_addr));
    this.addr += 4;
    this.Log(LogLevel.Info, "Reading addr {0:X} return value:{1:X}", this.addr - 4, result);
}
else {
    this.Log(LogLevel.Info, "[cpu: {0:X}] Reading Double word from: {1:X}, {2:X}.", pc, offset, result);
}
return result;
```

动态仿真环境搭建



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成功运行bootloader, 进入内核, 获得敏感信息

```
[ 0.000000] Kernel command line: rdinit=/usr/sbin/ runtime_start mtdoops.mtddev=mtdoops quiet alloc_snaps
hot trace_buf_size=5M rcutree.kthread_prio=80 uio_pdrv_genirq.of_id=generic-uio audit=1  RUNTIME_EXPECT_SUCC
ESS=true blkdevparts=mmcblk0:0x100000@0x00000000(BOOTFIP_0),0x100000@0x100000(BOOTFIP_1),0x100000@0x200000(BO
OTFIP_2),0x100000@0x300000(BOOTFIP_3),0x80000@0x400000(BOOTTERM1),0x80000@0x500000(BOOTTERM2),0x100000@0x6000
00(BOOT_A_0),0x100000@0x700000(BOOT_B_0),0x100000@0x800000(BOOT_A_1),0x100000@0x900000(BOOT_B_1),0x100000@0xA
00000(UBOOT_TERM1),0x100000@0xB00000(UBOOT_TERM2),0x50000@0xFB0000( ),0x180000@0x100000(KERNEL_A),0x8000
00@0x280000(CONFIG_A),0x180000@0x300000(KERNEL_B),0x800000@0x480000(CONFIG_B),0x180000@0x500000(SX_A),0
x180000@0x680000(SX_B),0x20000@0xF30000(VERSION_INFO_A),0x20000@0xF50000(VERSION_INFO_B),0x20000@0xF70000(S
ECRETS_A),0x20000@0xF90000(SECRETS_B),0x30000@0xF00000(MTD00PS),0x93D1C00@0x800000(EDR),0x2000000@0x113D1C00
(DISH_CONFIG) earlycon
```

内核启动参数

获得二进制设备树

```
## Loading fdt from FIT Image at a2000000 ...
Using 'rev2_proto2' configuration
Trying 'rev2_proto2_fdt' fdt subimage
Description: rev2 proto 2 device tree
Created: 2022-10-05 0:28:31 UTC
Type: Flat Device Tree
Compression: uncompressed
Data Start: 0xa2468a00
Data Size: 59749 Bytes = 58.3 KiB
Architecture: AArch64
Load Address: 0x8f000000
Verifying Hash Integrity ... OK
Loading fdt from 0xa2468a00 to 0x8f000000
Bootimg using the fdt blob at 0x8f000000
Uncompressing Kernel Image
Loading Ramdisk to 8f832000, end 8ffff93e ... OK
Loading Device Tree to 000000008f820000, end 000000008f831964 ... OK
```

```
/dts-v1/;
/ {
    #address-cells = <0x02>;
    #size-cells = <0x01>;
    compatible = "st,gllcfft";
    model = "spacex satellite user terminal";
    sx-dts-commit = "eda9413cf0e1ffbf71f5d30f33d83e4b786c6808";
    sx-dts-branch = "ssh://git@stash:7999/fswplat/linux.git master";
    sx-dts-build = "649c8a9c2e6650fb373de548092dd9280454e2dc";

    clocks {
        #address-cells = <0x02>;
        #size-cells = <0x01>;
        ranges;

        clk-sysin {
            #clock-cells = <0x00>;
            compatible = "fixed-clock";
            clock-frequency = <0x3938700>;
            phandle = <0x01>;
        };

        clk-unused {
            #clock-cells = <0x00>;
            compatible = "fixed-clock";
            clock-frequency = <0x00>;
            clock-output-names = "clk-unused";
            phandle = <0x05>;
        };
    };
};
```

反编译后的设备树

使用qemu-system跑起文件系统

```
./qemu-system-aarch64 \
-nographic \
-no-reboot \
-machine virt \
-machine type=virt \
-cpu cortex-a53 \
-smp 4 \
-m 16G \
-kernel "Image" \
-initrd ./rootfs.cpio.gz \
-dtb ./FDTs/custom.dtb \
-device e1000,netdev=vm0 \
-netdev tap,id=vm0,script=./networking/tap0-up.sh,downscript=./networking/tap0-down.sh \
-device e1000,netdev=vm1 \
-netdev tap,id=vm1,script=./networking/tap1-up.sh,downscript=./networking/tap1-down.sh \
-device e1000,netdev=vm2 \
-netdev tap,id=vm2,script=./networking/tap2-up.sh,downscript=./networking/tap2-down.sh \
-serial mon:stdio \
--append "$bootargs"
```

Qemu-system启动参数

```
ifconfig $1 0.0.0.0 promisc up
brctl addbr virbr1
brctl addif virbr1 $1

ip link set dev virbr1 up
ip link set dev $1 up

ip addr add 192.168.100.2/24 dev virbr1
echo 1 > /proc/sys/net/ipv4/ip_forward
```

网络配置脚本

但是仍然有很多专有程序因为硬件依赖问题

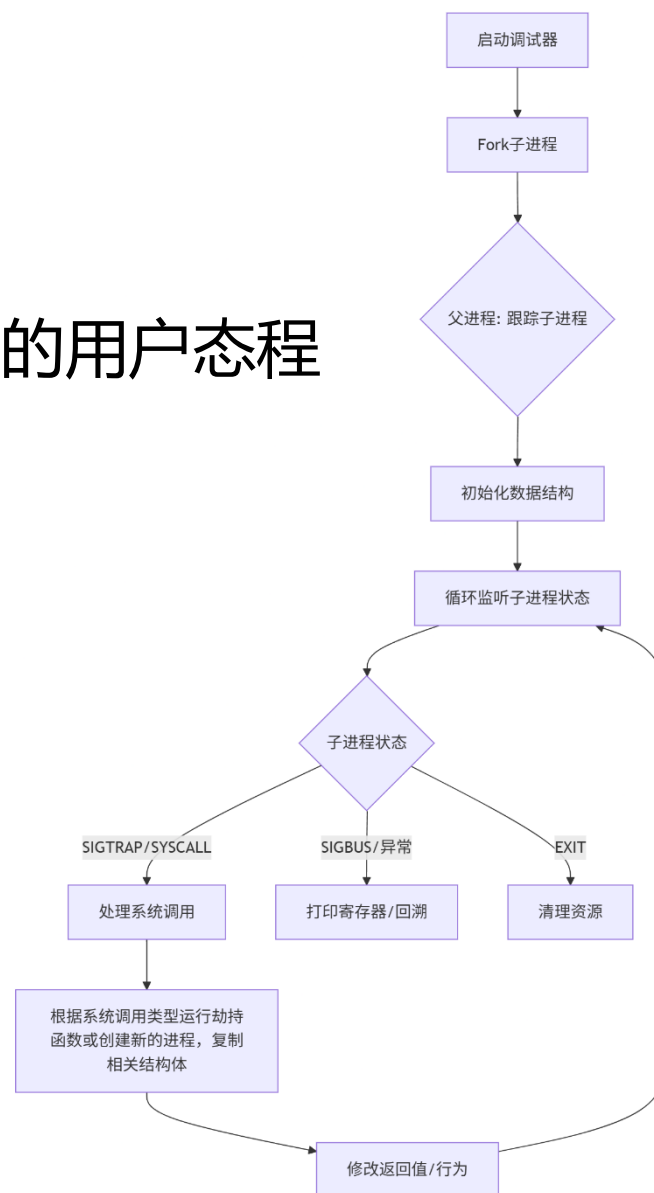
```
[root@unset runtime]# /sx/local/runtime/bin/ut_packet_pipeline --port=9001 --metrics_endpoint=0.0.0.0:9101 --prod_tunnel_root_ca=/sx/local/runtime/secrets/network/identity-certs.pem --dev_tunnel_root_ca=/sx/local/runtime/secrets/network/identity-certs.pem --identity_provider=/run/identity_provider --identity_key=/sx/local/runtime/secrets/network/identity-key.pem --identity_cert=/sx/local/runtime/secrets/network/identity-certs.pem --stsafe_cert=/sx/local/runtime/secrets/network/identity-certs.pem --stsafe_cipher=/run/stsafe_cipher --stsafe_cli=/sx/local/runtime/bin/stsafe_cli
metrics_singletons.so not loaded. Using local Metrics singletons.
This is expected for binaries that do not dynamically load shared libraries.
I0529 07:03:37.222247 728 init_spacex.cc:124] /sx/local/runtime/bin/ut_packet_pipeline is starting.
E0529 07:03:40.394528 728 stack_trace.cc:63] Process shut down by unexpected exception in thread 728
E0529 07:03:40.396699 728 stack_trace.cc:74] St13runtime_error: INTERNAL: Posix call failed: No such file or directory
E0529 07:03:40.397967 728 stack_trace.cc:80] Aborting
```

缺少相关硬件文件，因此无法打开

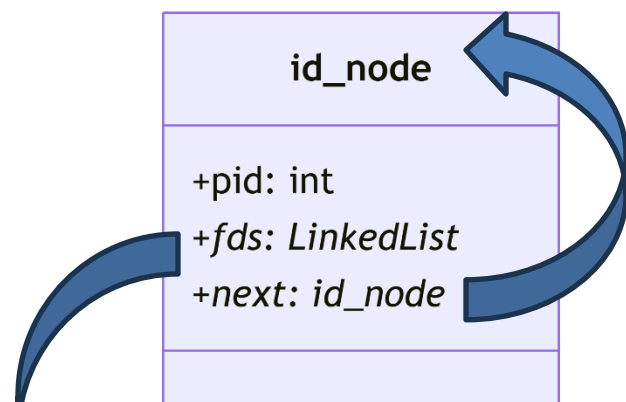
动态仿真环境搭建

基于ptrace的动态仿真工具

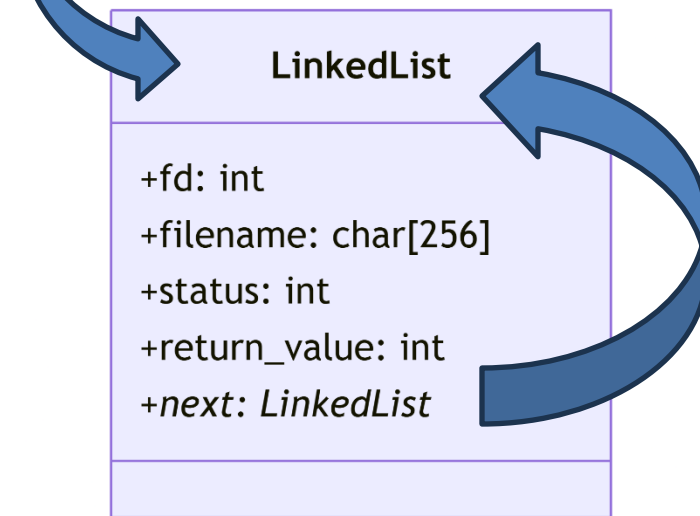
用于快速的仿真因硬件依赖无法仿真的用户态程序



关键数据结构



记录进程信息



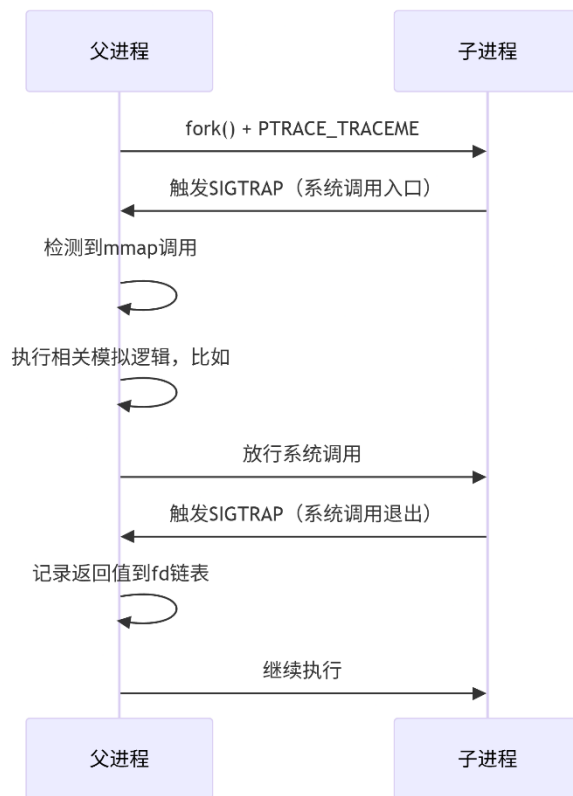
记录描述符信息

动态仿真环境搭建

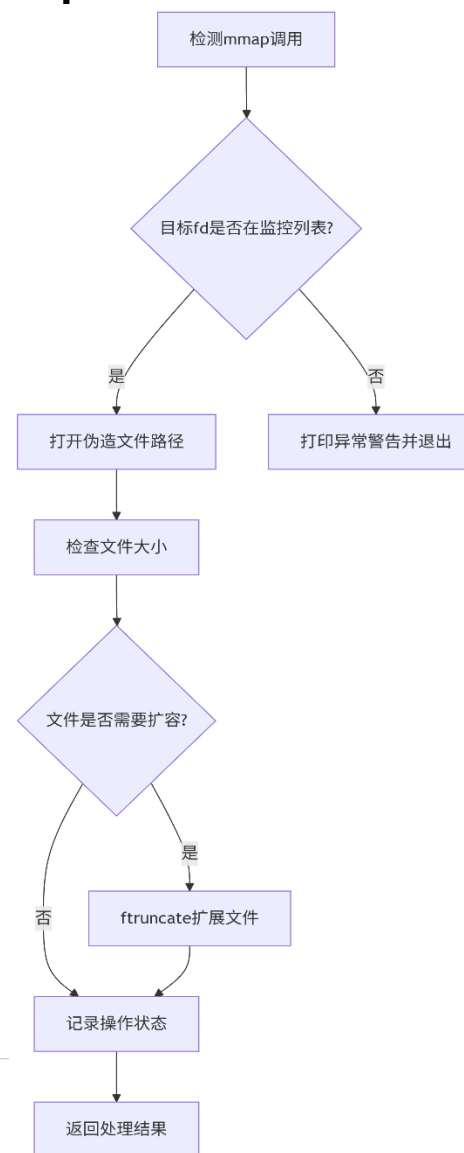


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Openat相关逻辑



mmap相关模拟逻辑



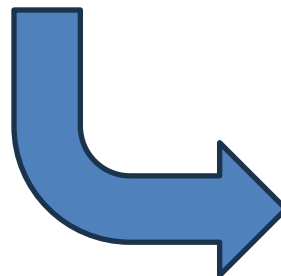
动态仿真环境搭建



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原先无法正常运行、初始化

```
[root@localhost ~]# ./bin/ut_packet_pipeline --port=9001 --metrics_endpoint=0.0.0.0:9101 --prod_tunnel_root_ca=/etc/ssl/certs/network/identity-certs.pem --dev_tunnel_root_ca=/etc/ssl/certs/network/identity-certs.pem --identity_provider=/run/identity_provider --identity_key=/etc/ssl/certs/network/identity-key.pem --identity_cert=/etc/ssl/certs/network/identity-certs.pem --stsafe_cert=/etc/ssl/certs/network/identity-certs.pem --stsafe_cipher=/run/stsafe_cipher --stsafe_cli=/run/stsafe_cli
metrics_singletons.so not loaded. Using local Metrics singletons.
This is expected for binaries that do not dynamically load shared libraries.
[890] [2024-04-10 08:55:51.735] [info] [space] [cc:12] [ut_packet_pipeline] is starting.
[890] [2024-04-10 08:55:51.735] [error] [space] [cc:63] Process shut down by unexpected exception in thread 735
[890] [2024-04-10 08:55:51.735] [error] [space] [cc:74] St13runtime_error: INTERNAL: Posix call failed: No such file or directory
[890] [2024-04-10 08:55:51.735] [error] [space] [cc:74] Aborting
Aborted
```



正常运行、初始化，进入了监听状态

```
[890] /proc/self/comm receive complete!
[890] Closing access to /proc/self/comm 3
[890] openat called with pathname: /dev/mem
[890] Action:9 for /dev/mem
[890] fd: 3 for /dev/mem
[890] Detected mmap read from /dev/mem with size 0x1000 at 0xc243000!
[890] PROT_FLAGS: PROT_READ | PROT_WRITE
[890] MAP_FLAGS: MAP_SHARED
[890] /tmp/dm1 Current file size: 16777216 bytes
[890] File expanded to 203702272 bytes
[890] Action:1 for /dev/mem
[890] return value: 7f91b17000
[890] Syscall number: 215
[890] Closing access to /dev/mem 3
[890] Syscall number: 94
[890] exited with status 0
[875] Syscall number: 260
[875] dup /usr/bin/config-tcam is reading!
[875] Action:10 for dup /usr/bin/config-tcam
[875] dup /usr/bin/config-tcam receive complete!
[875] Syscall number: 94
[875] exited with status 0
[747] Descriptors still blocked:
[747] Epoll_524288_966 is waiting for : eventfd_EFD_NONBLOCK | EFD_CLOEXEC_695 To ::70e2:67e3:7f00:0:8090
[746] Epoll_524288_966 is waiting for : eventfd_EFD_NONBLOCK | EFD_CLOEXEC_695 To ::70e2:67e3:7f00:0:8090
^Ia
```

部分逆向成果



基于模拟环境逆向开机启动项得到各个专有程序的具体用法，并根据其大致行为大致逆向出每个程序的大致用途

```
/bin/sh /local/runtime/bin/setup_ut.sh |builtin-logger setup_ut
/bin/sh /local/runtime/bin/avs.sh |builtin-logger avs
/bin/sh /local/runtime/bin/setup_multicast_ut_gw.sh |builtin-logger setup_multicast
/bin/sh /local/runtime/bin/mount_verity.sh /dev/blk/current/version_info /mnt/version_info /device-tree/security/dm-verity-pubkey
/bin/sh /local/runtime/bin/setup_security_packet_pipeline.sh
/bin/micropython /local/runtime/bin/setup_swupdate_keys.py
/usr/bin/httpd -f -u www-data -c /local/runtime/dat/common/httpd.conf |builtin-logger httpd
/bin/sh /local/runtime/bin/udhcpd_start.sh |builtin-logger udhcpd
/usr/bin/dnsd -i 192.168.100.1 -c /local/runtime/dat/common/dnsd.conf -U _packet |builtin-logger dnsd
/local/runtime/bin/system_housekeeping_monitor --env=dev --place=hq --role=user --inst=1 --string= --
config_dirs=board_rev:fused_state --devices_config=file_backed_devices --sharer_config=system_housekeeping_monitor_to_control --
sharer_service=system_housekeeping_monitor_to_control --telemetry_config=telemetry_housekeeping --disable_time_mark=true --
monitor_eth_phy_interface=eth0 --monitor_eth_phy_period=2
/local/runtime/bin/file_to_device_mapper --env=dev --place=hq --role=user --inst=1 --string= --config_dirs=board_rev:fused_state --
sharer_period=0.02 --sharer_service=file_to_device_mapper_to_control
/local/runtime/bin/phyfw --role=user --inst=1 --uid=sx_mac
/local/runtime/bin/rx_lmac --role=user --inst=1 --uid=sx_mac
/local/runtime/bin/tx_lmac --role=user --inst=1 --uid=sx_mac
/local/runtime/bin/umac --role=user --inst=1 --uid=sx_mac
/local/runtime/bin/user_terminal_control --env=dev --place=hq --role=user --inst=1 --config_dirs=board_rev:fused_state
/local/runtime/bin/unlock_service
/local/runtime/bin/analyze_alignment_server --role=user --inst=1
/local/runtime/bin/connection_manager --identity_provider=/run/identity_provider --identity_key=/local/runtime/secrets/network/
identity-key.pem --identity_cert=/local/runtime/secrets/network/identity-certs.pem --stsafes_uuid=/run/stsafes_uuid
/local/runtime/bin/ut_packet_pipeline_privileged |builtin-logger ut_packet_pipeline_privileged
/local/runtime/bin/ut_packet_pipeline --port=9001 --metrics_endpoint=0.0.0.0:9101 --prod_tunnel_root_ca=dat/common/
tunnel_ca_prod.pem --dev_tunnel_root_ca=dat/common/tunnel_ca_dev.pem --identity_provider=/run/identity_provider --identity_key=/
local/runtime/secrets/network/identity-key.pem --identity_cert=/local/runtime/secrets/network/identity-certs.pem --stsafes_cert=/
run/stsafes_leaf.pem --stsafes_cipher=/run/stsafes_cipher --stsafes_cli=/local/runtime/bin/stsafes_cli
```

使用grpc_cli在静态分析发现反射协议后动态分析得到grpc控制协议格式

```
luheng@luheng-virtual-machine:~/Desktop$ grpc_cli type 192.168.100.1:9200 .API.Device.Request
message Request {
  uint64 id = 1 [json_name = "id"];
  uint64 epoch_id = 14 [json_name = "epochId"];
  string target_id = 13 [json_name = "targetId"];
  oneof request {
    .API.Device.SignedData signed_request = 15 [json_name = "signedRequest"];
    .API.Device.GetNextIdRequest get_next_id = 1006 [json_name = "getNextId"];
    .API.Device.AuthenticateRequest authenticate = 1005 [json_name = "authenticate"];
    .API.Device.EnableFlowRequest enable_flow = 1018 [json_name = "enableFlow"];
    .API.Device.FactoryResetRequest factory_reset = 1011 [json_name = "factoryReset"];
    .API.Device.GetDeviceInfoRequest get_device_info = 1008 [json_name = "getDeviceInfo"];
    .API.Device.GetHistoryRequest get_history = 1007 [json_name = "getHistory"];
    .API.Device.GetLogRequest get_log = 1012 [json_name = "getLog"];
    .API.Device.GetNetworkInterfacesRequest get_network_interfaces = 1015 [json_name = "getNetworkInterfaces"];
    .API.Device.GetPingRequest get_ping = 1009 [json_name = "getPing"];
    .API.Device.PingHostRequest ping_host = 1016 [json_name = "pingHost"];
    .API.Device.GetStatusRequest get_status = 1004 [json_name = "getStatus"];
    .API.Device.RebootRequest reboot = 1001 [json_name = "reboot"];
    .API.Device.SetSkuRequest set_sku = 1013 [json_name = "setSku"];
    .API.Device.SetTrustedKeysRequest set_trusted_keys = 1010 [json_name = "setTrustedKeys"];
    .API.Device.SpeedTestRequest speed_test = 1003 [json_name = "speedTest"];
    .API.Device.UpdateRequest update = 1014 [json_name = "update"];
    .API.Device.GetLocationRequest get_location = 1017 [json_name = "getLocation"];
    .API.Device.GetHeapDumpRequest get_heap_dump = 1019 [json_name = "getHeapDump"];
    .API.Device.RestartControlRequest restart_control = 1020 [json_name = "restartControl"];
    .API.Device.FuseRequest fuse = 1021 [json_name = "fuse"];
    .API.Device.GetPersistentStatsRequest get_persistent_stats = 1022 [json_name = "getPersistentStats"];
    .API.Device.GetConnectionsRequest get_connections = 1023 [json_name = "getConnections"];
    .API.Device.FlushTelemRequest flush_telem = 1026 [json_name = "flushTelem"];
    .API.Device.StartSpeedtestRequest start_speedtest = 1027 [json_name = "startSpeedtest"];
    .API.Device.GetSpeedtestStatusRequest get_speedtest_status = 1028 [json_name = "getSpeedtestStatus"];
```

本研究成功开发了基于QEMU和ptrace的星地通信终端仿真平台实现了关键外设模拟和动态行为分析，初步逆向获得了启动参数、服务功能等核心信息。

由于时间限制，目前仅完成基础仿真环境搭建，尚未实现全部硬件仿真和全部系统程序逆向，但为后续研究提供了有效的技术路径和工具基础。

The background features three large, solid blue geometric shapes in the corners: a triangle in the top right, a triangle in the bottom right, and a larger triangle in the bottom left.

谢谢各位老师
请老师们批评指正